OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) IV-Semester (MAIN) Examinations Jun/Jul-2026

DESIGN AND ANALYSIS OF ALGORITHMS

me : 3 hours **Max. Marks : 60**

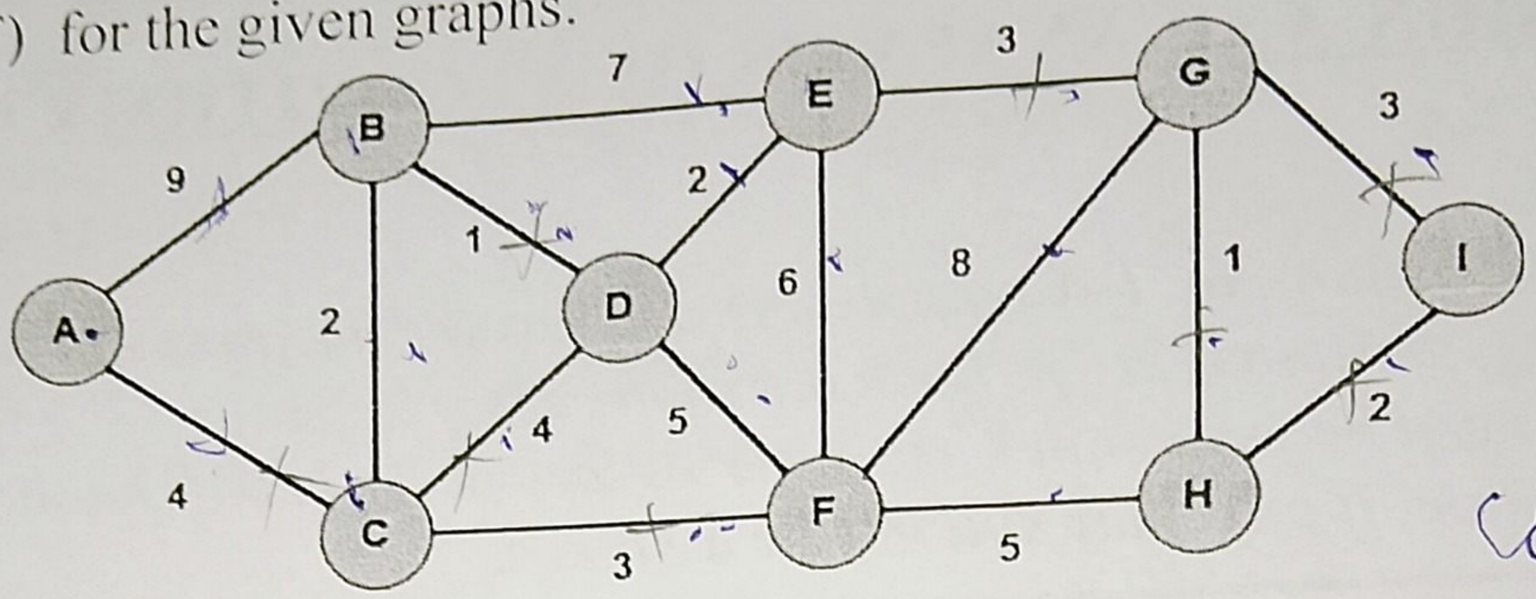
ote : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

			Marks	BT	C
1.	a)	Write the general control abstraction used in the Divide and Conquer strategy.	2	2	2
	b)	Why are recurrence relations used in the analysis of algorithms?	2	2	
	c)	Define augmenting path in a residual network and demonstrate it with a suitable example.	2	2	
	d)	State Cook's Theorem and describe its role in computational complexity theory.	2	2	
	e)	List the advantages of randomized algorithms.	2	1	
	f)	Determine the time complexity of the recurrence relation: $T(n) = T(n/2) + 1$ using substitution method.	2	3	
2.	a)	State and explain Master's Theorem in detail with examples for all cases.	6	2	
	b)	Discuss the various characteristics of algorithms and illustrate those using appropriate examples.	6	2	
3.	a)	A set of 5 jobs J_i along with their deadlines $D_i = (5, 3, 3, 2, 1)$ and profits $P_i = (200, 180, 190, 300, 120)$. Using the Greedy Job Sequencing with Deadlines algorithm, determine the optimal sequence of jobs that maximizes the total profit. Show the scheduling process and calculate the maximum profit obtained.	5	3	
	b)	Discuss the Travelling Salesman Problem and explain its solution using Branch and Bound.	7	4	

{P.T.O.}

4. a) Explain Dijkstra's algorithm for shortest path problem with an example. 6 3 2

b) Use Kruskal's algorithm to construct the Minimum Spanning Tree (MST) for the given graphs. 6 3 2



5. a) Prove that 3-SAT problem is NP-Complete. 6 4

b) Compare complexity classes P and NP in terms of problem-solving approaches and examples. 6 4

6. a) Explain randomized quicksort algorithm and analyze its complexity. 6 4

b) Explain approximation algorithms with suitable examples. 6 2

7. a) For a 0/1 Knapsack problem with n=4 items and knapsack capacity m=19, where the weights are (10, 15, 6, 9) and the profits are (2, 5, 8, 1), determine the optimal solution using the Merge and Purge technique. 7 3

b) Discuss the relationship between time complexity and space complexity in the context of time–space tradeoff. 5 4

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